clock cycles, RISC processors deliver over two 64-bit floating point results within a single clock cycle. They also have out-of-order instruction execution and a larger processor-to-memory bandwidth. All of these make the various RISC-based processors very attractive. The ASCI Q is housed at the Los Alamos National Laboratory, New Mexico, USA. It uses 4,096 EV-68 Alpha processors, each running at 1.25 MHz. ASCI Q is currently ranked as the world's second fastest supercomputer. IBM's processors also rank pretty high up this list, with ASCI White (ranked fourth) running on Power 3 processors, as well as Seaborg (ranked fifth) that runs on the Power 3+ processors.

Vector processors: Vector processors are a unique class of processors that work on entire vectors within a single instruction. Many intensive engineering and mathematical computational tasks require operations to be carried out on various elements of data. To solve such problems efficiently, vector processors were developed that can directly manipulate entire vectors of data. A vector processor executes fewer instructions, but manipulates an entire matrix or array of data in parallel.

For calculations that depend on working and sifting through large datasets, vector processors can be specialised to extract maximum performance; however, they are more expensive to design and produce. Today's HPC machine designers tend to use off-the-shelf RISC chips. Still, vector processors form a niche category and are certainly not phased out yet. Cray, for instance, is continuing work on its next generation vector processor; the SV-2; Japanese electronics giant NEC is doing so too.

NEC designed the fastest supercomputer of today—the Earth Simulator—made up of 5,120 vector CPUs. AMD and Intel are also very much in the reckoning, even as they come out with their 64-bit CPUs. HP has whole-heartedly adopted the Itanium 2 in their newer Superdome series, and so have SGI and NEC. The 32-bit Pentium 4 has already been enthusiastically welcomed by the cluster community. With its higher clock frequency, the 64-bit AMD Opteron is now being considered for SMP-style clusters.

Super software

As with all computers, supercomputers too depend on their operating system to provide basic resource management and scheduling features. Most supercomputers run some flavour of Unix. Those based on IBM RS/6000, as well as their P690 family of servers, work with IBM's Unix-based operating system—AIX. The Earth Simulator runs Super-UX, another Unix-based OS. The HP Superdome is pretty versatile, running HP-UX (a Unix based OS), Linux and even the Data Center edition of Windows Server 2003.

What's just as interesting are the high-level languages that such systems can work with. The choice is typically varied, with C, C++ and FORTRAN. For each such parallel computing system, compilers are specifically optimised to be able to extract maximum performance.

For DM machines, specific packages have been developed that will allow for rapid process management and message passing between nodes. The commonly used message passing standards are Parallel Virtual Machine (PVM), which is public domain software developed at the Oak Ridge National Laboratory. PVM comes with C and FORTRAN libraries, to take care of process management, as well as to allow for message passing between processes. Message Passing Interface, MPI, is another such standard. This is also public domain software, and has been developed at the Argonne National Laboratory, Illinois, USA.

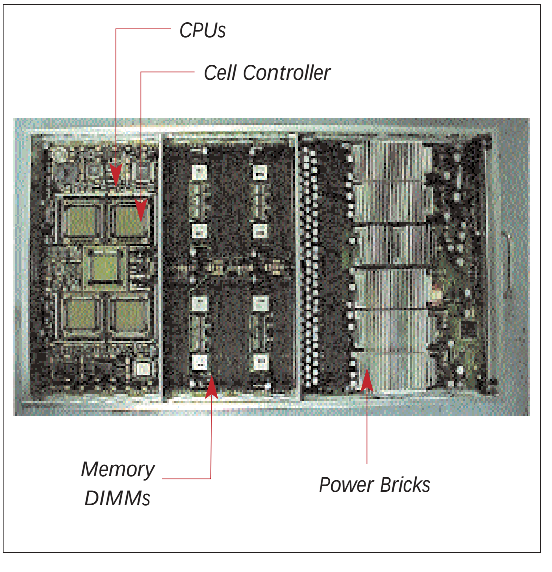
Supercomputing Performance

Computer performance is measured on the basis of Flops (Floating Point Operations Per second). The machine is benchmarked using standard tools, and this value is used as a measure of the possible performance of the machine. The Earth Simulator located in Japan, has a maximum speed of 35,860 GFlops. To get an idea of how big this figure is—a Pentium 4 can perform 4 double precision Flops per clock cycle. So a Pentium 4, running at a clock speed of 2.4 GHz, can have a theoretical maximum performance equivalent of 9.6 GFlop—practically, though, you'll never reach this speed.

C-DAC's Param Padma cluster runs at 532 GFlops. The slowest supercomputer on the top 500 list is an old HP Superdome 750 series-based computer with a compute speed of 245 GFlops.

Superdome, not your everyday PC

We looked at a typical high performance computing machine that is on the list of the top 500 supercomputers in the world (***www.top500.org***)—the HP Superdome. Physically, the HP Superdome 9000 stands only about a foot higher than a 360 litre refrigerator. The basic unit of the Superdome is a cell that's made up of four processors, modelled on a symmetric multiprocessing architecture—the cell controller ASIC (Application Specific Integrated Circuit), the SD-RAM memory modules, the data buses and the power units. A cell processor can be either the HP PA-8700 or an Intel Itanium 2. The cell controller ASIC determines and co-ordinates memory access between the components on the cell board, and monitors request the need to communicate between other cells.



The Superdome cell—the core of the performance machine

There's also a memory controller ASIC, whose job is to co-ordinate memory requests between the cell controller and the memory modules. Individual cells are connected by a crossbar interconnect that supports up to eight cells; two crossbar backplanes can be linked together to support up to 16 cells.

The Superdome is a ccNUMA system, and provides an SMP model to the OS, by letting any processor access memory distributed anywhere in the system. The crossbar bandwidth for a 64-way (64- processor) Superdome works out to an impressive 64 GBps. In practice, the Superdome 875 has been certified at speeds of 530 GFlops. It can run HP-UX, Linux, or Windows Server 2003 Data Center edition. It comes bundled